Pattulingam K

Software Quality Analyst II · Quality engineering, test automation and AI-assisted testing

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9+
Years in quality engineering

2
Domains: insurance and reinsurance

6
AI-assisted tools built

ABOUT

Software Quality Analyst II with 9+ years testing life, annuity and reinsurance platforms. I build Playwright and TypeScript automation, validate SOAP and web-service integrations, and use LLM-based tools to generate test assets and remove repetitive manual checks.

I work on TPP, a new business and underwriting platform that takes an application from case import through requirements management and underwriting to policy issue. My focus is keeping those end-to-end flows reliable release after release, using automation to catch regressions early and clear test design to cover the business rules.

I also build AI-assisted tools around the testing work itself: generators that turn JIRA stories, scope documents and estimates into test cases and plans, and validators for DTCC and NSCC files and carrier PDF forms. The goal is less repetitive checking, faster feedback and fewer defects reaching production.

Role: Software Quality Analyst II

Since: May 2017

Domains: Life, annuities, reinsurance

Based in: Coimbatore, India

SKILLS

Expert

Playwright with TypeScript · End-to-end test automation · Data-driven test automation · Regression and integration testing · API testing with SoapUI and Postman · Test design from user stories and acceptance criteria · Defect management

Proficient

AI-assisted test design and scripting · SQL Server · JIRA · TestRail · Azure DevOps · WildFly (Linux and Windows) · iHub · Smoke testing

Growing

LLM prompting and agent workflows · Python · JavaScript · Flask and FastAPI · JIRA and Confluence APIs · PyMuPDF · JUnit

EXPERIENCE

Software Quality Analyst II

Zinnia · May 2017 to present

- Build and maintain Playwright and TypeScript automation for critical end-to-end business workflows, extending the suite with each release to widen regression coverage and speed up release cycles.
- Built an end-to-end Playwright and TypeScript flow for annuity contract creation in the policy administration system: 20 steps from product selection through payment, exchange and replacement, and cash batch release, driven by Excel test data and producing Excel execution reports.
- Built a client-specific AI agent that generates test cases and integration test plans from JIRA stories, scope documents and estimates, in the client's standard plan format.
- Document each change request as a reusable QA reference guide: scope summary, source-to-target mapping, key tables and fields, a SQL query library for selecting and validating test data, business rules, validation strategy and a defect template. Five guides so far, being consolidated into a QA handbook for future projects.
- Received the Deliver Value award under Values in Action for building AI tools that reduce day-to-day manual activity.

- Use AI-assisted tools across the testing cycle: test case creation, automation script development and debugging.
- Turn JIRA user stories, business requirements and acceptance criteria into test plans, scenarios and cases.
- Validate SOAP integrations with SoapUI and JUnit and web services with Postman; test system interfaces and add new products through batch processing.
- Deploy builds to WildFly and confirm they are stable before testing.
- Raise and track defects through to verification, and advise stakeholders during sprint and enhancement discussions based on test results.

American Family Life Insurance

TPP Runtime for life insurance: case import, case management, requirements management and underwriting on one platform, used by new business users, service users and underwriters.

General Reinsurance Corporation, US

TPP Runtime for reinsurance new business and underwriting, used by data ingestion specialists, underwriters, medical directors and administrative staff.

Playwright · TypeScript · SoapUI · Postman · SQL Server · WildFly · JIRA · TestRail · Azure DevOps · LLM tooling

EDUCATION

Bachelor of Engineering, Electrical & Electronics Engineering

Anna University · 2011 to 2014

CERTIFICATIONS

Learn Playwright with TypeScript (Web & API Testing)

Udemy · July 2026 · 70 hours · Instructor: Pavan Kumar

PROJECTS

Tools and automation I built to replace repetitive manual checks in insurance and annuity workflows. The file tools run locally, in the browser or on localhost, and never upload files.

AI FOR QA

Client-specific AI test case generator

An LLM-powered agent that generates test cases and integration test plans for a specific insurance client, following that client's change-order process and standard plan format. It reads a JIRA story with its acceptance criteria, comments, screenshots and linked Confluence documents. For change requests it combines the scope document and the estimate: the estimate sets scenarios and case counts, the scope sets behavior and expected results, and conflicts between the two are flagged for review. Output goes to Excel, JSON or Markdown; the saved plan can be edited and rebuilt without another model call. Works with OpenAI or Anthropic models and ships with a Cursor skill.

Python · LLM agents · OpenAI / Anthropic · JIRA API · Confluence API · Pydantic · Typer · openpyxl

TEST AUTOMATION

End-to-end contract creation automation

An end-to-end Playwright and TypeScript test that creates an annuity contract in a policy administration system in 20 steps: company and product selection, contract amount, annuitant and joint owner details, beneficiaries, optional pages, writing and servicing agents, validation-error handling (retry and ignore dialogs, SSN correction), payment application, exchange and replacement set-up, cash batch creation and release, application processing and pending payment. Test data is read from Excel, so scenarios change without code edits, and a reporting utility writes a timestamped Excel report with a summary, step-by-step results and pass-rate statistics. A shorter 12-step data-entry check gives quick feedback in about three minutes. Runs on system Chrome with traces captured on retry.

Playwright · TypeScript · Data-driven testing · Excel (SheetJS) · Excel reporting

ANNUITIES

DTCC field mapper

Maps a fixed-width DTCC APP/SUB transaction file to a master dictionary workbook, validates every field against the workbook code lists and exports a corrected file. One Application Control Number edit updates every record. Runs as a shareable single-folder web app.

JavaScript · SheetJS · Python · FastAPI

ANNUITIES

NSCC POV file validator

Validates Positions Focused, New and Full Refresh files against the NSCC layout, applying record-level and cross-record business rules, and lets users browse errors by contract before submission.

JavaScript · Python · openpyxl

SETTLEMENT

STL contract field parser

A single-file browser tool that turns NSCC STL fixed-length files into readable per-contract breakdowns, resolves coded values to descriptions and exports to CSV or Excel.

JavaScript · HTML · SheetJS

FORMS

PDF forms updater

Fills both fillable and flat carrier PDF forms using per-form overlay templates, including checkmarks and signature fields, and downloads the completed PDF.

Python · Flask · PyMuPDF

FORMS

Annuity application PDF editor

Extracts 246 fields from a flat fixed index annuity application, lets users edit them by section and writes an updated PDF in the original layout.

Python · Flask · PyMuPDF

ACCOMPLISHMENTS

VALUES IN ACTION

Deliver Value award

Recognized for building AI tools that cut day-to-day manual effort for the team, including test case and test plan generation and validators for industry file formats.

AI-assisted tools · Productivity · Quality engineering